

Genetic Analysis of Chemosensory Traits in Human Twins

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We explored genetic influences on the perception of taste and smell stimuli. Adult twins rated the chemosensory aspects of water, sucrose, sodium chloride, citric acid, ethanol, quinine hydrochloride, phenylthiocarbamide (PTC), potassium chloride, calcium chloride, cinnamon, androstenone, Galaxolide™, cilantro, and basil. For most traits, individual differences were stable over time and some traits were heritable (h^2 from 0.41 to 0.71). Subjects were genotyped for 44 single nucleotide polymorphisms within and near genes related to taste and smell. The results of these association analyses confirmed previous genotype–phenotype results for PTC, quinine, and androstenone. New associations were detected for ratings of basil and a bitter taste receptor gene, TAS2R60, and between cilantro and variants in three genes (TRPA1, GNAT3, and TAS2R50). The flavor of ethanol was related to variation within an olfactory receptor gene (OR7D4) and a gene encoding a subunit of the epithelial sodium channel (SCNN1D). Our study demonstrates that person-to-person differences in the taste and smell perception of simple foods and drinks are partially accounted for by genetic variation within chemosensory pathways.